Examples

Example 2.6 In IEEE double precision arithmetic the associative rule for summation and multiplication may be violated as follows. The computation of

$$1 + 10^{17} - 10^{17}$$

as

$$1 \oplus (10^{17} \ominus 10^{17})$$

gives the correct answer 1, while the result of the machine summation

$$(1 \oplus 10^{17}) \ominus 10^{17}$$

is 0.

Why?

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$$1 = 0.1 \times 10^{1} = 0.00000000000000001 \times 10^{18}$$

$$0.1 \text{ x } 10^{18}$$

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In turn, the computation of

$$10^{155}10^{155}10^{-250}$$

as

$$10^{155} \otimes (10^{155} \otimes 10^{-250})$$

will produce an accurate approximation to the correct answer 10^{60}, while the attempt to compute

$$(10^{155} \otimes 10^{155}) \otimes 10^{-250}$$

will give an overflow.

One of the most dangerous operations during the computations is the subtraction of numbers that are very close to each other (although in most cases this is done exactly in machine arithmetic). In this case a *subtractive cancellation* may occur.

Example 2.7 Consider the numbers

$$x = 0.123456789\xi, \ y = 0.123456789\eta,$$

which agree in their first 9 significant decimal digits. The result of the subtraction is

$$z = x - y = \pm 0.000000000|\xi - \eta|.$$

If the original x and y are subject to errors, likely to be at least of order 10^{-10}, then the subtraction has brought these errors into prominence and z may be dominated by error. If in particular the first 9 digits of x and y are true, and the digits ξ, η are uncertain, then there may be no true digit in the result z.

Solving some illustrative problems by MATLAB

The commands

```
>> x = [-1:-1:-18]'; x = 10.^x
```

will produce the 18–vector $x = [10^{-1}, 10^{-2}, \ldots, 10^{-18}]^\top$. With this vector we may compute the quantities

$$y(k) = \frac{(1 + x(k)) - 1}{x(k)}, \ k = 1, 2, \ldots, 18,$$

where $x(k)$ are the elements of x, from

```
>> y = ((1 + x) - 1)./x
```

The result is

```
y =
 1.00000000000000
 1.00000000000000
 0.99999999999989
 0.99999999999989
 1.00000000000655
 0.99999999991773
 1.00000000058387
 0.00000000392253
 1.00000008274037
 1.00000008274037
 1.00000008274037
 1.00008890058234
 0.99920072216264
 0.99920072216264
 1.11022302462516
                0
                0
                0
```

Since the exact result is 1, we see that for $k = 15$ there is a relative error of 11 percent while for $k \geq 16$ the result is totally wrong with relative error of 100 percent.

The computation of the vector of ones

$$y(k) = \frac{(1 + x(k))^2 - 1 - 2x(k)}{x^2(k)}$$

by the command

```
>> y = ((1 + x).^2 - 1 - 2*x)./x.^2
```

will give

```
y =
 1.00000000000002
 1.00000000000006
 0.99999999969742
 0.99999999171889
 1.00000139299870
 0.99992436739695
 1.01087806660425
-1.215...
 ...
```

We see that for $k = 7$ the result is computed by 11 percent relative error, while for $k \geq 8$ the result is completely useless (for $k = 8$ even the sign is wrong and the relative error is 221 percent).

Example 2.8 If we compute

$$y = \frac{(1+x)^2 - (2x+1)}{x^2}, \ x \neq 0,$$

in the ANSI/IEEE 754-1985 Standard for

$$x = 10^{-1}, 10^{-2}, \ldots, 10^{-n}, \ldots,$$

we shall see that for $n \geq 7$ the computed result $\widehat{y}$ is far from the correct answer $y = 1$; even negative values for $\widehat{y}$ will appear. The reason for this phenomenon is that we have cancellation in the numerator for small x. For $n > 8$ the result of the subtraction is of order $\varepsilon \approx 10^{-16}$ or less and there is no correct digit in the computed numerator. To worsen the situation, this wrong result is divided by a denominator that is less than 10^{-16}. This example is very instructive because the input of order 10^{-8} and an output 1 are by no means close to the boundaries of the standard range $[10^{-324}, 10^{308}]$ of $\mathbb{M}$.

Similarly, the computation of the quantities

$$y(k) = \frac{(1+x(k))^3 - 1 - 3x(k) - 3x^2(k)}{x^3(k)},$$

all equal to 1, by the corresponding command gives

```
y =
    1.00000000000035
    0.99999999991242
    0.99999950547556
    0.99987350828302
    1.10953927987086
-202.14...
  ...
```

The results are self-evident.

Otherwise speaking, *cancellation is the loss of true significant digits when small numbers are computed by subtraction from large numbers.*

Often subtractive cancellation can be avoided by a simple reformulation of the problem.

Example 2.10 The expression

$$y = \sqrt{1+x} - 1$$

may be computed as

$$y = \frac{x}{1+\sqrt{1+x}}$$

to avoid cancellation for small x.

As a summary, one has to remember the following practical rules about machine arithmetic.

- *The rounding unit* $\mathbf{u}$ *is about* 10^{-16} *which means that the rounding and the arithmetic operations are done with 15-16 true significant decimal digits.*

- *For safe and accurate arithmetic calculations the (non-zero) operands and the result must have absolute values in the normal range between* 10^{-308} *and* 10^{308}*. A tolerance of several orders of magnitude may increase the reliability of computations.*

- *Subtraction of close approximate numbers should be avoided whenever possible or at least be carefully watched.*

Control of Numerical Errors - Guidelines

- Avoid subtracting two equal numbers, to prevent loss of significance
- Rearrange or reformulate the problem to avoid subtractive cancellation.
- When adding or subtracting numbers, sort the numbers and work with the smallest numbers first (to avoid loss of significance)
- Prediction of total numerical error is very complicated for even moderately sized problems.
- Repeat the computations with different parameters and compare the results.

Control of Numerical Errors - Guidelines

- **Sensitivity analysis**: see how your solution changes when you change the model parameters or input values.
- Try different numerical algorithms (with different convergence and stability properties)
- Critical problems involving human life:
  - Let two or more independent groups solve it, and compare the results.